George Emil Sadek

AI/ML Engineer | Computer Vision | Applied AI Backend

Professional Summary

Final-year Computer Engineering student building deployable AI systems across computer vision and applied ML backends. Hands-on with PyTorch, TensorFlow, OpenCV, FastAPI, Flask, Docker, and SQL across image classification, segmentation, video understanding, and perception workflows. Comfortable taking models from notebooks to inference APIs, containers, and lightweight web interfaces.

Technical Skills

Languages: Python, SQL, C++
ML / CV: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, CNNs, image segmentation, video classification, model evaluation
Backend / Deployment: FastAPI, Flask, Django, REST APIs, Docker, Git, Linux
Data / Tools: Pandas, NumPy, MySQL, SQL Server, SQLite, Streamlit, ROS 2, CARLA

Experience

Computer Vision Intern - Cellula Technologies — Jan 2026 - Apr 2026

- Built end-to-end computer vision pipelines in TensorFlow and PyTorch across medical image classification, multispectral satellite segmentation, and surveillance video analysis.
- Developed and integrated production-style inference workflows using FastAPI, Django, Streamlit, Docker, and geospatial I/O for model serving and visualization.
- Applied practical deep learning methods including transfer learning, mixed precision, augmentation, threshold tuning, data cleaning, and evaluation beyond accuracy to improve model reliability.

Intern - Elevvo Internship Program — Aug 2025 - Sep 2025

- Built machine learning projects across regression, clustering, classification, and deep learning using reproducible notebook-based workflows on real datasets.
- Applied preprocessing, feature engineering, scaling, imbalance handling, model comparison, and performance evaluation across supervised and unsupervised learning tasks.
- Delivered clearly documented experiments and implementation outputs through structured notebooks, technical reporting, and version-controlled project organization.

Selected Projects

Shoplifting Detection from Surveillance Video [GitHub] — *PyTorch, R3D-18, Django*

- Built a video classification pipeline for shoplifting detection using an R3D-18 backbone; the best experiment reached F1 = 0.9867 on the cleaned test split.
- Added a Django demo [GitHub] for video upload, inference, and result preview.

Flood Rapid Mapping / Water Segmentation API [GitHub] — *FastAPI, Docker, PyTorch*

- Rebuilt a production-style segmentation service for flood-area mapping from optical imagery around a DeepChannelAdapter + UnetPlusPlus(EfficientNet-B4) model.
- Exposed endpoints for masks, probability maps, visualization, and GeoTIFF export, then packaged the service with Docker.

Dental Image Classification [GitHub] — *TensorFlow, CNNs, Medical Imaging*

- Trained a 7-class dental image classifier with custom CNN and transfer-learning baselines; the best run reached 99.81% test accuracy on the project dataset.

Paper Piano with Computer Vision [GitHub] — *OpenCV, Python, Real-time Vision*

- Developed a real-time webcam application that detects a hand-drawn piano, tracks finger touches, and maps interactions to piano-note playback with calibration controls.

Education

B.Sc. in Computer Engineering, Faculty of Engineering at Shoubra, Benha University — Sep 2021 - Jun 2026

CGPA: 3.4 / 4.0

Awards & Additional Information

Shell Eco-marathon Autonomous Programming Competition - 2nd Place

Perception Engineer

- Worked on camera-LiDAR sensor fusion and perception pipeline development for the autonomous vehicle stack.
- Contributed to environment understanding workflows supporting downstream autonomy tasks in simulation.

Certifications: Machine Learning Specialization, Deep Learning Specialization, AWS Machine Learning Foundations.